Ivey Davis, PhD

They/them

davis@astron.nl https://github.com/iveydavis iveydavis.info

Exo-solar space weather from a multi-wavelength perspective
Star-planet interactions
Magneto- and plasmasphere processes

Academic Positions

Bell Burnell Postdoctoral Fellow
The Netherlands Institute for Radio Astronomy (ASTRON) *Nov. 2025-Present*

Education

PhD., M.Sc. of Astrophysics @ California Institute of Technology *2020 - 2025*
Advisor: Prof. Gregg Hallinan
Thesis: *A Portrait of the Sun as a Young Star: Studying Space Weather Around Young Suns in the Optical and Radio*
Highlights: Development of a high-precision, ground based optical observatory. Identifying physical limitations to plasma emission process. Identifying radio counterpart to stellar superflares.

B.Sc. of Astronomy @ University of New Mexico *2016 - 2020*
Advisors: Prof. Greg Taylor & Prof. Rouzbeh Allahverdi
Honors project: Early kinetic decoupling of dark matter
Additional work: development of beamformer observing and calibration method for the Long Wavelength Array

Science Leadership and Affiliations

Flarescope *2020 - Present*
Project scientist for the optical photometry instrument Flarescope, operating at sub-mmag precision. My role included **designing** the observatory and optical path, **installing hardware**, **automating** observatory operations, and **developing pipelines** for calibration, reduction, and photometry.

Owens Valley Radio Observatory Long Wavelength Array (OVRO-LWA) *2020 - Present*
Commissioner for the beamformed observing mode of the OVRO-LWA. This involved regularly **conducting observations**, **developing a framework** for observation scheduling, **defining standards** for data-recorder products, and **producing data-reduction pipelines** for the beamformed data.

Keck Institute for Space Science: Observing Stellar and Exoplanet Particle Environments *2023 - 2025*
Lead for the transient particles working group, a part of a larger effort to consolidate community understanding of stellar/exoplanetary particle environments. I **coordinated an international group** of astronomers and synthesized their input into a chapter describing both familiar and novel transient particle detection methods for the Sun and stars.

Publications

First author:

5. **I. Davis**, et al, "Summary of the First Year of the Space Weather Around Young Suns Program: 900 Hours of Low-frequency Radio and Optical Data Dedicated to Young, Solar-type Stars", Accepted for publication in ApJ, 2026

4. **I. Davis**, et al, "Detection of Radio Emission from Super-flaring Solar-Type Stars in the VLA Sky Survey", ApJ 2026

3. **I. Davis**, et al, "A dedicated system for coordinated radio and optical monitoring of the space weather of young, solar-type stars", ApJ, 2025
2. **I. Davis**, et al,"Rapid low-frequency spectral evolution of WX UMa", A&A, 2021
1. **Ivey Davis**, Greg B. Taylor, & Jayce Dowell, "Observing Flare Stars Below 100 MHz with the LWA", MNRAS, 2020

Co-lead author:

1. Loyd, R. O. P., **et al.** "The Exospace Weather Frontier Final Report for the Keck Institute for Space Studies", KISS, 2025

Talks and Workshops

11. COSPAR 2026	**Invited** talk, *Aug. 2026*
10. UA High-contrast Imaging of Exoplanets	**Invited** talk, *Mar. 2025*
9. Tuesday UVa NRAO Astronomy Lunch Talk	**Invited** talk, *Nov. 2024*
8. Stars in Brisbane	Contributed talk, *Nov. 2024*
7. Laboratory For Atmospheric and Space Physics Seminar	**Invited** talk, *Oct. 2024*
6. Radio Stars in the Era of New Observatories	Contributed talk, *Apr. 2024*
5. AAS 243	Contributed talk 355.08, *Jan. 2024*
4. KISS "Stellar and Exoplanet Particle Environments" Workshop	*Nov. 2023*
3. Palomar Science Meeting 2023	Contributed talk, *Jun. 2023*
2. AAS 241	Contributed talk 346.07, *Jan. 2023*
1. AAS 240	Contributed talk 107.03, *Jun. 2022*

Skills

Instrument design and commissioning

Experienced in **design**, **construction**, and **calibration** of optical observatories. Familiar with **performing** stress tests, **diagnosing** failure modes, and **developing automated-observing pipelines** for both radio and optical observatories.

Data reduction and analysis

Experienced in **source finding and tracking** in images with novel PSFs and unstable atmos-/ionosphere (code available here), **extracting** light curves and **identifying** flare structures (`flare_id`). Additionally experienced with radio astronomy tools (CASA, wsclean, AIPS), **direct manipulation** of visibilities for imaging, visibility-based beamforming, and sky subtraction and de-dispersion of dynamic spectra.

Familiar instruments/data products: VLA [S and P band visibilities], LOFAR [HBA and LBA visibilities], (OVRO-)LWA [visibilities and beamformed], TESS [light curves], RACS [images], Flarescope [images]

Observing

Completed ≈**1000 hours** of all-night observing coordinated between Flarescope and OVRO-LWA. This included developing and executing all-night observing plans and managing observatory start up/shutdown/contingencies for both observatories.

Modelling

Developed a package for modelling stellar coronae and predicting the **shape of of stellar type II and III bursts** (`swabs`). Additionally developing a package for predicting the **contrast of stellar CMEs in direct-imaging observations** (`cme_imaging`).

Communication

Active in public outreach in the form of giving talks, facilitating public observing nights, and participating in physics demonstration shows. I also served as an **Astrobites** author, writing summaries of journal articles to make them more accessible to undergraduate students.

Successful Proposals

Observing

PI for VLA proposal 25B-319 *2025*

"Constraining T Tauri Coherent Emission Through Multi-Observatory Coordination"

Co-PI for ATCA proposal C3786 *2025*

'Multiwavelength constraints on the origin of coherent radio bursts from T-Tauri stars"

PI for VLA proposal 20B-252 *2020*

"Observing the EQ Pegasi System with the Expanded Long Wavelength Array"

Funding

Co-I JPL President's and Director's Research and Development Fund (**$750k**) *2025*

"Extrasolar Space Weather Monitoring with Coordinated Optical and Low-Frequency Radio Observations"

Co-I JPL Researchers on Campus Fund (**$46k**) *2021, 2022*

"Real time milli-mag differential photometry of stars with Flarescope"

Mentorship and Service

Referee for ApJ papers *2025 - Ongoing*

Advising students

Bramuel Nakholi *Summer 2026*

ASTRON Summer Student Programme

Project: *Largest search for contemporaneous stellar radio and optical emission from LOFAR and TESS*

Klara Matuszewska *Summer 2024*

Caltech Summer Undergraduate Research Fellow (SURF)

Project: *"Building a Stellar Radio Catalog of Sources With TESS Flares"*.

Awarded the Vodopia-Hasson Poster Prize.

ASTRON Postdoc Representative *2026 - Ongoing*

Responsible for communicating goals and concerns of ASTRON postdocs to the broader Dutch postdoc community and representing early-career researchers' interests to the NWO, union bodies, and institute administration.

Astrobites Author *Jan. 2023 - September 2025*

Wrote summaries of astronomy articles for various academic levels and served on the policy committee to identify societal goals of the astronomical community. Selected writings:

A Sleepy Grad's Guide to Healthier Observing

Making Sense of Convoluted Images

A Rad(iation) Belt Around an Ultracool Dwarf

Caltech Gender Minorities and Women in PMA *2021-2024*

Served on the **organizing committee** and as **president**. I organized events on campus relating to gender minorities and women in physics, math, and astronomy (PMA); this included running journal clubs related to the intersection of identity and science, running social events to connect the broader gender-minority community in PMA, and helping facilitate a mentorship program.

Caltech Respect as Part of Research *2022-2024*

Facilitated and **organized** a yearly seminar for incoming Caltech PMA graduate students. I both trained other volunteers and facilitated peer-led, identity-based harassment and assault prevention discussions.

Caltech Astronomy graduate student representative *Fall 2023-Fall 2024*

Organized regular meetings for astronomy and physics graduate students to delegate department tasks and address concerns. Facilitated discussion between students and faculty.

Caltech PMA Graduate Student Advisory Board *Fall 2022- Fall 2023*

Served on a board of graduate students who met with PMA administation and chairs to discuss graduate students' concerns and advocate on their behalf.

Public Outreach

Caltech Stargazing Lecture *May 2025*

Sequoia National Park Dark Skies Festival Lecture *Sept. 2024*

iTelescope Webinar *Dec. 2023*

Caltech Annular Eclipse Viewing *Oct. 2023*

Organized a public outreach event to observe the partial annular eclipse from Caltech that was attended by $\sim$1,000 people. I also gave a public lecture on solar radio science at Caltech.

Astro on Tap: Pasadena *Sept. 2022*

Owens Valley Radio Observatory Fall Lecture Series *Nov. 2021*